

Applied Industrial Organization

ClassWork 1

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1. Create a Word file named `yourname_yoursurname_cw1.docx`
2. Create a new do-file
3. Set the working directory
4. Save the do-file as `yourname_yoursurname_cw1.do`
5. Load the online database named “census2” (by doing `webuse census2`)
6. Find an appropriate graph to show the distribution of the variable `region`. Save and insert it in the Word file.
7. See the relation between the two *categorical* variables `age` and `region`.
8. Look at the options of the command used in the previous point and see if there is a way to test the independence of the two variables (tip: Pearson). Copy and briefly comment on the output.
9. Now, you are interested in understanding if there is a relation between the *categorical variable* `region` and the *continuous variable* `pop`. Find an appropriate graph to describe this relation. Save and insert it in the Word file.
10. Now, you want to understand if, in places in which there are more marriages, there are also more divorces. Find an appropriate graph to describe this relation. Save and insert it in the Word file.
11. The relation shown in the previous point may simply depend on the fact that in places where more people live, there are more marriages and more divorces. Compute the ratio of both variables and population and see if the relation changes. Save and insert it in the Word file.
12. It seems that there is an anomalous value (*outlier*). Find a way to remove it from the graph. Save and insert it in the Word file.

13. Create a new variable, which is the median value of the age across states and within regions.
14. Generate a variable that takes value 1 if the age of the country is higher or equal than its own median.
15. Keep only observations for which the dummy computed in the previous point is equal to one.
16. Re-run the graph produced in point 11. Save and insert it in the Word file.

Remaks:

- If you need to copy-paste an output from Stata to a Word file, you can do that by following these steps:
 1. Select the output that you want to copy.
 2. Then right-click and Copy.
 3. Paste it in Word.
 4. Select the output and change the font in **Courier New**
- Pay attention to the layout of the graphs (e.g. title, axis titles, etc.)